Linear and Logistic Regression

A birth-weight analysis: odds ratios and Box–Cox transformations

Statistical Inference 2026–27
Lecture Notes

1 Begin with the data and the medical questions

We have information on 189 births recorded at Baystate Medical Center, Springfield, Massachusetts, in 1986. The dataset is available directly in R as `MASS::birthwt`. We will follow one analysis from the observed data to the fitted models and then to a choice of response scale.

Question 1. How is smoking during pregnancy associated with the probability that a baby has low birth weight?

Question 2. How is smoking associated with the baby's actual birth weight, measured in grams?

These questions use the same births but different responses. For the first, we use a binary indicator. For the second, we retain the continuous measurement from which that indicator was constructed. Maternal age and maternal weight will be additional predictors in both models. The observations are treated as independent. This is an observational analysis: the fitted smoking coefficients describe associations, not automatically causal effects.

Variable	Meaning
`bwt`	Infant birth weight in grams: the continuous response.
`low`	1 if birth weight is below 2500 g; 0 otherwise: the binary response.
`smoke`	Smoking during pregnancy; recoded as No/Yes.
`age`	Maternal age in years.
`lwt`	Maternal weight at the last menstrual period, in pounds.

```
library(MASS)
d <- MASS::birthwt
d$smoke <- factor(d$smoke, c(0, 1), c("No", "Yes"))
head(d[, c("low", "bwt", "smoke", "age", "lwt")])
```

Output: the first six observations.

low	bwt	smoke	age	lwt
0	2523	No	19	182
0	2551	No	33	155
0	2557	Yes	20	105
0	2594	Yes	21	108
0	2600	Yes	18	107
0	2622	No	21	124

For example, the first baby weighs 2523 g, just above the threshold, so `low`=0. The first six rows all have `low`=0; they are not representative of the full response distribution. We must use all 189 observations to form the group comparison. We must also keep the two weight variables distinct: `bwt` is the infant's weight in grams, while `lwt` is maternal weight in pounds.

2 Use the data to understand an odds ratio

2.1 First calculate probabilities

```
with(d, table(smoke, low))
```

Smoking	Not low: `low`=0	Low: `low`=1	Total
No	86	29	115
Yes	44	30	74

Among smokers, 30 of 74 infants have low birth weight. Among nonsmokers, 29 of 115 do. Their observed event probabilities are therefore

$$\widehat{p}_1 = \frac{30}{74} = 0.4054, \qquad \widehat{p}_0 = \frac{29}{115} = 0.2522.$$

In words: about 40.5% of infants in the smoking group and 25.2% in the nonsmoking group have low birth weight. Probability divides the number of events by the total number of observations in the group.

2.2 Now change the denominator: from probability to odds

Odds compare the number of events with the number of *nonevents*. In the smoking group, the odds are 30/44, not 30/74:

$$\widehat{o}_1 = \frac{\widehat{p}_1}{1-\widehat{p}_1} = \frac{30/74}{44/74} = \frac{30}{44} = 0.6818.$$

There are about 0.68 low-birth-weight births per birth that is not low weight. This is not a 68% probability. For nonsmokers, the odds are

$$\widehat{o}_0 = \frac{29}{86} = 0.3372.$$

The general conversion is $o = p/(1-p)$. To go back, solve $o(1-p) = p$, obtaining $p = o/(1+o)$. For instance, odds of 2 mean two events per nonevent, which corresponds to probability 2/3.

2.3 Compare the two sets of odds

Using nonsmokers as the reference group, the odds ratio is

$$\boxed{\widehat{\mathrm{OR}} = \frac{\widehat{o}_1}{\widehat{o}_0} = \frac{30/44}{29/86} = \frac{30 \times 86}{44 \times 29} = 2.0219.}$$

Thus, in these data, the odds of low birth weight among smokers are about twice the odds among nonsmokers. Always identify both the event and the reference group when interpreting an OR. Reversing the groups gives the reciprocal OR, approximately 0.4946.

Do not replace "odds" by "probability." Here the risk ratio and risk difference are

$$\widehat{\mathrm{RR}} = \frac{30/74}{29/115} = 1.6076, \qquad \widehat{\mathrm{RD}} = \frac{30}{74} - \frac{29}{115} = 0.1532.$$

The same table therefore gives about twice the odds, about 61% higher risk, and a 15.3-percentage-point risk difference. These statements use different measures; they do not contradict one another.

An OR alone does not tell us the event probability. If the reference probability is p_0 and the OR is θ, then

$$\frac{p_1}{1-p_1} = \theta \frac{p_0}{1-p_0} \quad \Longrightarrow \quad p_1 = \frac{\theta p_0}{1 - p_0 + \theta p_0}.$$

For OR $= 2$, baseline risk 0.10 gives risk 0.1818, while baseline risk 0.50 gives risk 0.6667. This is why a reported odds ratio should not be interpreted as an absolute increase in the chance of an event.

3 Fit logistic regression to the binary response

3.1 What changes when we move beyond the table?

The two-group table compares smokers and nonsmokers as observed. Those groups may also differ in maternal age and weight. Logistic regression allows us to compare their event probabilities at the *same* specified values of these predictors.

Let $Y_i = \texttt{low}_i$, and let S_i equal 1 for a smoker and 0 otherwise. Define $A_i = \texttt{age}_i - 25$ and $W_i = (\texttt{lwt}_i - 130)/10$. Centering gives a useful reference profile; dividing weight by ten makes its coefficient describe a ten-pound increase. Neither operation loses information.

For independent observations conditional on the predictors, specify

$$Y_i \mid S_i, A_i, W_i \sim \mathrm{Bernoulli}(p_i), \qquad \log \frac{p_i}{1-p_i} = \beta_0 + \beta_S S_i + \beta_A A_i + \beta_W W_i.$$

The left-hand side is the *log odds of the conditional event probability.* We do not take logs of the observed zeros and ones. Writing the right-hand side as η_i, the fitted probability is $p_i = e^{\eta_i}/(1+e^{\eta_i})$, which is always between zero and one.

R estimates the coefficients by maximizing the Bernoulli log likelihood

$$\ell(\beta) = \sum_i \{y_i \eta_i - \log(1 + e^{\eta_i})\}.$$

The following commands fit the model and print its coefficient table.

```
d$age_c <- d$age - 25
d$lwt10 <- (d$lwt - 130)/10
logit_fit <- glm(low ~ smoke + age_c + lwt10,
                 family = binomial, data = d)
coef(summary(logit_fit))
```

Term	Estimate	Std. Error	z value	$\Pr(>\|z\|)$
(Intercept)	-1.184650	0.230859	-5.1315	2.87×10^{-7}
smokeYes	0.670764	0.325878	2.0583	0.03956
age_c	-0.038995	0.032726	-1.1915	0.23344
lwt10	-0.121385	0.061349	-1.9786	0.04786

3.2 Read the smoking row across all four columns

Estimate: 0.670764. At the same maternal age and weight, the fitted log odds for a smoker exceed those for a nonsmoker by 0.670764. This number is not a probability difference. To express the comparison as an odds ratio, exponentiate:

$$\widehat{\mathrm{OR}}_{\mathrm{smoking}} = e^{0.670764} = 1.9557.$$

The adjusted odds are about 95.6% higher among smokers under this model. Why exponentiate? The fitted odds for a smoker and a nonsmoker at the same A, W are respectively $e^{\beta_0+\beta_S+\beta_A A+\beta_W W}$ and $e^{\beta_0+\beta_A A+\beta_W W}$. Their ratio cancels every shared term and leaves e^{β_S}.

Std. Error: 0.325878. This estimates the sampling uncertainty of the fitted *log-odds coefficient*. It is on the same scale as the estimate. It is not the standard deviation of the binary outcomes and is not itself the standard error of the odds ratio. In logistic regression the usual coefficient covariance estimate is the inverse fitted information matrix; R reports square roots of its diagonal entries.

z value: 2.0583. R divides the estimate by its standard error:

$$z = \frac{0.670764}{0.325878} = 2.0583.$$

For the null coefficient zero, the fitted estimate is about 2.06 estimated standard errors above zero. A zero smoking coefficient corresponds to conditional $\mathrm{OR} = e^0 = 1$.

p-value: 0.03956. Under that null and the usual large-sample likelihood assumptions, the statistic is approximately standard normal. The reported two-sided p-value is

$$2\{1 - \Phi(|2.0583|)\} = 0.03956.$$

It measures how unusual a statistic at least this far from zero would be under the null and model assumptions. It is not the probability that smoking has no association. At a conventional 5% level this row provides evidence against a zero conditional coefficient; the OR gives the size and direction of the estimated association.

3.3 Read the other rows in the context of their units

- **Intercept.** At $S = A = W = 0$, the mother is a nonsmoker, age 25, weighing 130 lb. The fitted log odds are -1.184650 and the odds are $e^{-1.184650} = 0.30585$. Converting odds to probability gives $0.30585/(1 + 0.30585) = 0.2342$. The intercept's p-value tests log odds zero, equivalently baseline probability 1/2; it does not assess the overall usefulness of the regression.
- **Age.** One additional year multiplies the fitted odds by $e^{-0.038995} = 0.9618$, holding smoking and maternal weight fixed. The p-value 0.2334 does not provide strong evidence against a zero age

coefficient in this particular model. It does not establish that age is irrelevant or has exactly no association.

- **Maternal weight.** Ten additional pounds multiply the fitted odds by $e^{-0.121385} = 0.8857$: about 11.4% lower odds, holding smoking and age fixed. Twenty additional pounds give the multiplier $e^{2(-0.121385)} \approx 0.7844$. The p-value 0.0479 is close to 0.05; it should not be read as a sharp distinction between an important and an unimportant predictor.

Why z, not t? The ordinary binomial logistic model has a variance determined by its probability: $\mathrm{Var}(Y_i \mid x_i) = p_i(1 - p_i)$. The standard coefficient table uses asymptotic normal, or z, calibration. It does not have the exact Student t result from a Gaussian linear model with an estimated error variance. A logistic table should not contain an additional t statistic for the same row.

3.4 Convert the fitted model back to event probabilities

```
nd <- data.frame(smoke = factor(c("No", "Yes"),
                 levels = levels(d$smoke)), age_c = 0, lwt10 = 0)
predict(logit_fit, nd, type = "response")
```

At age 25 and weight 130 lb	Fitted log odds	Fitted probability
Nonsmoker	-1.18465	0.2342
Smoker	$-1.18465 + 0.67076 = -0.51389$	0.3743

At this profile, the fitted risk difference is about 14.0 percentage points and the risk ratio is about 1.60. The OR remains 1.956. Without interactions the model gives the same smoking OR at each age and weight, but the probability difference changes because the baseline risk changes. The adjusted OR need not equal the table's OR 2.022: one is a conditional model comparison, while the other compares the observed groups overall.

4 Use linear regression for actual birth weight

4.1 Change the response, keeping the predictors

Now ask about birth weight in grams rather than the chance of crossing the 2500 g threshold. Set $B_i = \texttt{bwt}_i$ and fit

$$B_i = \alpha_0 + \alpha_S S_i + \alpha_A A_i + \alpha_W W_i + \varepsilon_i.$$

The conditional mean is the linear predictor when $\mathbb{E}(\varepsilon_i \mid x_i) = 0$. The classical Gaussian model additionally assumes independent errors with common variance σ^2 and normal distributions. These are assumptions about variation *around the conditional means*, not about whether the predictors are normally distributed.

```
raw_fit <- lm(bwt ~ smoke + age_c + lwt10, data = d,
              y = TRUE, qr = TRUE)
coef(summary(raw_fit))
```

Term	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
(Intercept)	3063.35868	67.99372	45.0536	1.14×10^{-101}
smokeYes	−269.25698	105.80060	−2.5449	0.01175
age_c	7.15385	9.92452	0.7208	0.47193
lwt10	40.15515	17.19669	2.3351	0.02061

Additional output is residual standard deviation $s = 708.76$ g, residual degrees of freedom $189 - 4 = 185$, and $R^2 = 0.0704$.

4.2 Read the smoking row again: the scale has changed

Estimate: −269.26 **g.** At the same maternal age and weight, the fitted mean birth weight for a smoker is about 269 g lower than for a nonsmoker. This is now an additive difference in grams, not a log-odds difference. Exponentiating this coefficient would not produce an odds ratio or a useful interpretation.

Std. Error: 105.80 **g.** This measures uncertainty in the estimated adjusted mean difference. Its units are grams, matching the coefficient. It is different from the residual standard deviation 708.76 g, which measures individual outcome variation around the fitted conditional means.

t value and p-value. The ratio is

$$t = \frac{-269.25698}{105.80060} = -2.5449.$$

Under a zero coefficient and the Gaussian linear-model assumptions, this statistic has a t_{185} distribution. The two-sided p-value is $2\{1 - F_{t_{185}}(2.5449)\} = 0.01175$. The reference law is t because the unknown error variance is estimated from the residuals. Normality gives this exact finite-sample reference law; outside the model, the usual table is not automatically exact.

4.3 Interpret the complete model

The intercept 3063.36 g is the fitted mean for a nonsmoker aged 25 and weighing 130 lb. At that same age and weight, a smoker's fitted mean is $3063.36 - 269.26 = 2794.10$ g. A ten-pound increase in maternal weight adds 40.16 g to fitted mean birth weight, holding the other predictors fixed. A one-year increase in age adds 7.15 g, with substantial uncertainty relative to the estimate and p-value 0.4719.

The very small intercept p-value tests whether the baseline mean is zero grams, a scientifically uninteresting reference in this setting. It does not establish that the model predicts well. Indeed, $R^2 = 0.0704$ means that these predictors explain about 7% of the sample variation in birth weight. This is compatible with evidence for some associations and considerable remaining variation across individual births.

Compare the two smoking results. Logistic regression estimates about 1.956 times the odds of low birth weight; linear regression estimates about 269 g lower mean birth weight. The direction is coherent: lower birth weight can accompany a higher probability of falling below 2500 g. But the two models answer different questions, so their coefficients and p-values are not expected to agree numerically.

5 Why consider transforming the continuous response?

The linear model makes a particular choice of scale: effects add in *grams*, and the classical error model assumes the same conditional spread in grams across predictor profiles. Those assumptions can be inappropriate for some positive medical outcomes.

For example, suppose outcome variation is proportional to its typical level: patients with a typical value of 4000 might vary much more in absolute units than patients with a typical value of 2000. A common error standard deviation would not describe both well. Or suppose a predictor produces proportional changes, such as a 5% increase, rather than a fixed additive change. Such patterns motivate looking for a scale on which an additive model with more nearly constant spread is a useful approximation.

5.1 What a transformation changes

If we replace B by $\log B$, a doubling from 1000 to 2000 and a doubling from 2000 to 4000 both become an increment of $\log 2$. Large absolute gaps at high response levels are compressed. A square root compresses them less strongly. A power greater than one does the opposite: it expands differences at higher response levels. Thus a transformation changes both the mean relationship being modeled and the scale on which error variation is described.

What it does not do. Low R^2 alone is not evidence that a transformation is required. Transformation cannot supply omitted predictive information or turn observational associations into causal effects. Positivity alone is not a justification for taking logs. Likewise, a skewed pooled response histogram does not by itself show that the conditional regression errors are skewed.

Our question here is exploratory and specific: holding the predictors fixed, does a power transformation of positive birth weight give appreciably stronger support under a Gaussian regression working model? We have not established in advance that birth weight needs transformation. The analysis is allowed to recommend retaining grams. We apply this procedure to `bwt`, not to the binary response `low`. The logistic link already serves a different purpose: it maps a conditional probability to log odds.

6 Fit Box–Cox transformations and examine the output

6.1 Put the candidate scales into one family

For a strictly positive response B, define

$$T_\lambda(B) = \begin{cases} (B^\lambda - 1)/\lambda, & \lambda \neq 0, \\ \log B, & \lambda = 0. \end{cases}$$

The formula at zero is the continuous limit, because $B^\lambda = e^{\lambda \log B} = 1 + \lambda \log B + o(\lambda)$. The derivative $T'_\lambda(B) = B^{\lambda-1}$ is positive, so every candidate preserves the ordering of the observed birth weights.

Power	Transformation	Scale
1	$B - 1$	Original scale, shifted by one
1/2	$2(\sqrt{B} - 1)$	Square-root scale, rescaled and shifted
0	$\log B$	Logarithmic scale
-1	$1 - 1/B$	Negative reciprocal, shifted

Why is $\lambda = 1$ called "no transformation" if the result is $B - 1$? With an intercept, subtracting one from every response simply subtracts one from the intercept. The slopes and residuals are unchanged; adding one back recovers the ordinary fitted values in grams.

For each candidate λ, use the same regression predictors:

$$T_\lambda(B_i) = \alpha_0 + \alpha_S S_i + \alpha_A A_i + \alpha_W W_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

The coefficients and variance are re-estimated for each power. We are not taking the original fitted coefficients and raising them to a power.

6.2 How does the method compare different powers?

For a fixed power, let $z_{i,\lambda} = T_\lambda(B_i)$ and let RSS_λ be the residual sum of squares after regressing z_λ on the same design matrix X. We cannot just choose the smallest unadjusted RSS: changing the response scale changes the units and can mechanically shrink those numbers.

Instead, express each likelihood on the original birth-weight scale. The change of variables has derivative $B_i^{\lambda-1}$, so the log likelihood contains the corresponding Jacobian term:

$$\ell(\alpha, \sigma^2, \lambda) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{\sum_i \{T_\lambda(B_i) - x_i^T\alpha\}^2}{2\sigma^2} + (\lambda - 1)\sum_i \log B_i.$$

The last term makes the comparison use the same original-data scale. For fixed λ, maximizing over α is ordinary least squares, and maximizing over the variance gives $\widehat{\sigma}^2_\lambda = \mathrm{RSS}_\lambda/n$. The divisor is n here because this is likelihood maximization, not the unbiased residual-variance estimate used in a linear-model t table. Substitution gives the *profile log likelihood*

$$\boxed{\ell_p(\lambda) = -\frac{n}{2}\log\left(\frac{\mathrm{RSS}_\lambda}{n}\right) + (\lambda - 1)\sum_i \log B_i + C,}$$

where C does not depend on λ. "Profile" means that the other parameters have already been fitted as well as possible for each specified power. We now compare those optimized values.

Scope of this working model. For nonzero powers, the image of positive responses is bounded on one side: $T_\lambda(B) > -1/\lambda$ for positive λ, and $T_\lambda(B) < -1/\lambda$ for negative λ. An unrestricted Gaussian distribution extends beyond that range. The conventional Box–Cox calculation is therefore a Gaussian working approximation when such probability is negligible. A normalized truncated-normal model would need a different likelihood. The log case maps positive responses onto the whole real line.

6.3 Compute and plot the profile in R

```
bc <- MASS::boxcox(raw_fit, lambda = seq(-2, 3, .002),
                   plotit = FALSE, interp = FALSE)
lambda_hat <- bc$x[which.max(bc$y)]
cutoff <- max(bc$y) - qchisq(.95, 1)/2
lambda_ci <- range(bc$x[bc$y >= cutoff])
c(lambda_hat = lambda_hat, lower = lambda_ci[1],
  upper = lambda_ci[2])
```

Output:

```
lambda_hat      lower      upper
     1.310      0.880      1.768
```

The grid searches powers from -2 to 3. It gives a maximum at 1.310. The peak and interval endpoints are inside the search range, so the reported answers are not merely artifacts of stopping at a grid boundary. The endpoints are approximate to the grid resolution.

```
plot(bc$x, bc$y-max(bc$y), type = "l", lwd = 3,
     xlab = expression(lambda),
     ylab = "Profile log-likelihood minus maximum")
abline(h = -qchisq(.95, 1)/2, lty = 2)
abline(v = c(0, 1), lty = 3)
```

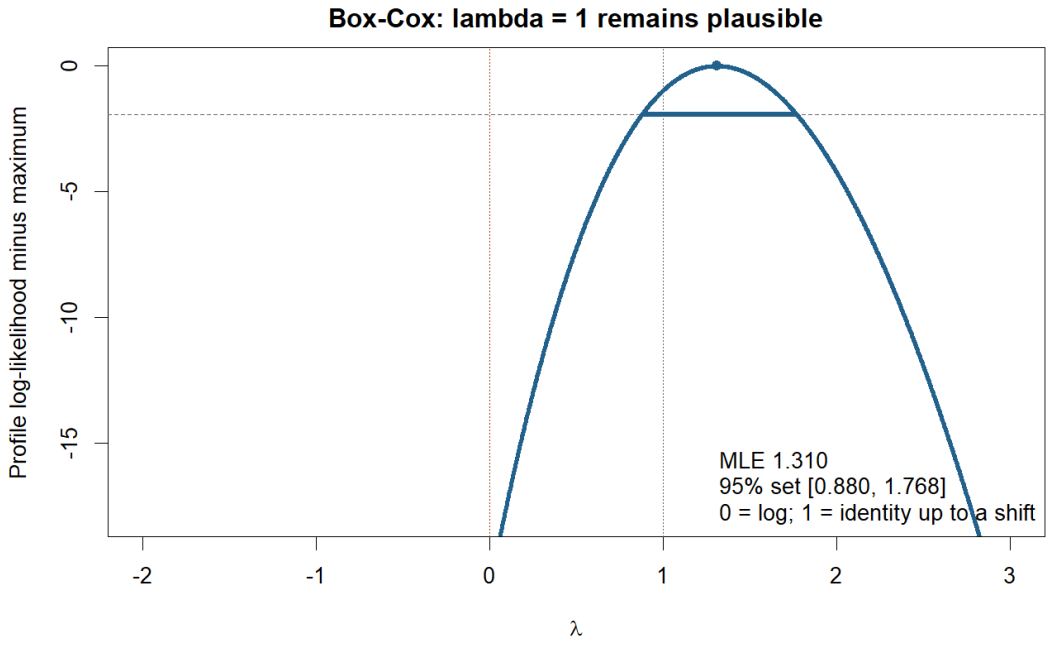


Figure 1: Profile log likelihood for the Box–Cox power. The displayed curve has its maximum shifted to zero. The horizontal dashed line is the approximate 95% likelihood-ratio cutoff; the thick segment marks the corresponding interval.

Read the horizontal axis first. Each point on the axis is a different transformation: 0 is log, 1 is identity, and 1.310 is the estimated power. Moving along this axis also refits the regression.

Read the vertical axis next. Higher means better optimized log likelihood. We subtract the maximum so the peak is zero; this changes neither the maximizing power nor any likelihood difference. The absolute vertical level is not meaningful for our comparison.

Read the dashed line last. The approximate 95% set contains the powers whose profile log likelihood is no more than $\chi^2_{1,.95}/2 = 1.9207$ below the peak. Equivalently,

$$2\{\ell_p(\widehat{\lambda}) - \ell_p(\lambda)\} \leq \chi^2_{1,.95}.$$

This is the usual one-parameter likelihood-ratio approximation under regularity conditions and the stated working model, not an exact finite-sample interval. It records uncertainty in the power rather than treating 1.310 as a uniquely established scale. We will return to the final choice after seeing what the fitted-power model predicts.

6.4 Fit the selected power, then undo the transformation

To see how a fitted-power model works, temporarily use the maximizing power 1.310. There are three operations, and their order matters.

First transform the observed birth weights:

$$z_i = \frac{B_i^{1.310} - 1}{1.310}.$$

Next regress these transformed observations on the predictors. This produces fitted values $\widehat{z}_i = x_i^T \widehat{\alpha}$ on the transformed scale. They are not in grams.

Finally solve backwards for an original-scale fitted value. Starting from $\widehat{z} = (\widetilde{B}^\lambda - 1)/\lambda$, we get

$$\lambda\widehat{z} = \widetilde{B}^\lambda - 1 \quad \Longrightarrow \quad \widetilde{B}^\lambda = 1 + \lambda\widehat{z} \quad \Longrightarrow \quad \boxed{\widetilde{B} = (1 + \lambda\widehat{z})^{1/\lambda}.}$$

The inverse at $\lambda = 0$ is $e^{\widehat{z}}$. For a nonzero power, the expression requires $1 + \lambda\widehat{z} > 0$. The code checks that condition rather than silently computing invalid powers.

```
# These expressions apply to the estimated nonzero power 1.310.
d$z <- (d$bwt^lambda_hat - 1)/lambda_hat
bc_fit <- lm(z ~ smoke + age_c + lwt10, data = d)
z_hat <- fitted(bc_fit)
stopifnot(all(1 + lambda_hat*z_hat > 0))
inverse_fitted <- (1 + lambda_hat*z_hat)^(1/lambda_hat)
plot(inverse_fitted, d$bwt,
     xlab = "Inverse fitted birth weight (g)",
     ylab = "Observed birth weight (g)")
abline(0, 1, lty = 2)
```

The separate full script includes a log-limit branch, numerically stable versions of these formulas, and the code for both plotted panels. It follows the same sequence without defining custom functions.

6.5 Do not confuse inverse fitted location with mean birth weight

A nonlinear inverse does not commute with averaging:

$$T_\lambda^{-1}\{\mathbb{E}[T_\lambda(B) \mid x]\} \neq \mathbb{E}(B \mid x) \quad \text{in general.}$$

Our inverse fitted curves estimate a transformed-scale location brought back to grams, not a bias-corrected arithmetic mean. For a strictly increasing transformation, conditional medians transform directly. Thus, if the transformed error has median zero and compatible support, the inverse location is the original-scale median. Because nonzero Box–Cox powers use the Gaussian working approximation described earlier, we label the plot conservatively as *inverse fitted values*.

The log case gives an exact illustration of the difference. If $\log B \mid x \sim N(\mu(x), \sigma^2)$, then

$$\operatorname{median}(B \mid x) = e^{\mu(x)}, \qquad \mathbb{E}(B \mid x) = e^{\mu(x)+\sigma^2/2}.$$

The mean needs an extra factor because $\mathbb{E}(e^{\varepsilon}) = e^{\sigma^2/2}$ for normal ε. For a general power, obtaining an original-scale mean likewise requires averaging the inverse over an appropriate conditional error distribution. Simply applying the inverse to a fitted transformed mean does not do that.

7 Make the final choice of power

We can now put the numerical and graphical evidence together. The estimated maximum is $\widehat{\lambda} = 1.310$, but the approximate 95% profile interval is $[0.880, 1.768]$. The identity power 1 lies inside this interval; the log power 0 does not. In log-likelihood units, the identity is only 0.978 below the maximum, while the log power is 20.784 below it. The cutoff used for the interval is 1.9207.

The inverse fitted curves also show that using the estimated power produces only modest changes to the fitted locations over the illustrated predictor range. They do not demonstrate a substantial practical gain that would compel us to abandon the interpretable gram scale.

Final choice for this example: retain $\lambda = 1$. With an intercept this is ordinary linear regression on birth weight, up to an irrelevant response shift. We therefore retain

$$\widehat{\mathbb{E}}(B \mid S, A, W) = 3063.36 - 269.26S + 7.15A + 40.16W \quad \text{grams}.$$

At age 25 and weight 130 lb, this gives fitted means 3063.36 g for nonsmokers and 2794.10 g for smokers. These are the predictions from the selected original-scale model, not the inverse fitted values of 3095.55 and 2823.01 g used to illustrate the estimated-power model.

```
# The profile supports the simple identity power in this example.
final_fit <- raw_fit
predict(final_fit, nd)
#       1        2
# 3063.359 2794.102
```

This choice does not prove that the Gaussian linear model is correct. It says that, within this fixed predictor specification and the Box–Cox comparison, the evidence does not justify giving up the simpler original scale. The important lesson is that a transformation method is allowed to conclude that transformation is unnecessary. We fitted the estimated power to understand its operation, inspected its implied original-scale fitted values, and then made the modeling choice using both uncertainty and interpretation.